# Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · JavaScript · AI/ML Integration · Scalable Platform Engineering · CI/CD · Production Reliability · OOP · AWS · Generative AI · Mentoring

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## PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years delivering scalable, maintainable production systems in complex, compliance-critical environments - with a track record of integrating AI and machine learning components that enhance real software capabilities. At Entrata, owned end-to-end delivery of 20+ production modules across 120+ Agile sprints: Python and JavaScript backends, OOP-based architecture, CI/CD-driven quality, and observability tooling reducing triage time 40% at 99.5% uptime for 100K+ monthly active users. Built and shipped two generative AI systems from iterative development to production: AgentProbe (open-source LLM evaluation platform with CI-integrated regression gating) and LeaseI (8-agent agentic AI system with RAG pipelines and HITL escalation). Conducted 300+ code reviews, mentored 5+ engineers, and consistently translated complex business problems into measurable, maintainable technical outcomes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**Engineering & Backend:** Python (8yr), JavaScript, TypeScript, SQL, FastAPI, PHP, Bash, OOP (object-oriented programming), Docker, Microservices, REST APIs, CI/CD (GitHub Actions, continuous integration), DevOps practices, AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, Linux, Git

**AI & Generative AI Integration:** LangGraph, LangChain, Agentic AI, RAG Pipelines, LLM Orchestration, Generative AI Tools, Model Evaluation, LLM-as-Judge, Hallucination Detection, HITL, Prompt Engineering, ChromaDB, OpenAI API, Anthropic API, Groq, MCP, BERT, spaCy, NLP

**Quality & Data:** Code Review, Debugging, Continuous Development, Performance Monitoring, Observability, Scalable System Design, PHI Compliance, scikit-learn, PyTorch, NumPy, Pandas, PySpark, Feature Engineering, Tableau, Agile/Scrum

## EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Integrated AI and machine learning components into a production agentic system (LangGraph, Python) - iterative development from requirements through deployment; continuous integration pipeline enforcing quality at every release.
- Leveraged generative AI tools to accelerate development; translated complex mission-planning business requirements into scalable, maintainable software with deterministic guardrails.

### George Mason University

Aug 2024 - May 2026

**Graduate Teaching Assistant & Research Associate**

- Built scalable ML pipelines (Python, scikit-learn, PyTorch) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end iterative development from data ingestion through model evaluation and statistical validation across 190+ countries.
- Mentored 600+ students in Python, OOP, SQL, and software engineering best practices; designed dashboards improving research comprehension by 40%.

### Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **Led delivery of scoped features through collaborative engineering teams** - 20+ scalable production modules across 120+ Agile sprints; translated complex business problems into measurable technical outcomes; continuous integration with daily deployment cadence; 99.5% uptime SLA at 100K+ monthly active users.
- **Integrated AI and machine learning components to enhance platform capabilities** - AI-driven observability and monitoring reducing debugging time 40%; ML-powered data validation across compliance-critical pipelines; Python and JavaScript OOP-based architecture designed for scalability and maintainability.
- **Code review, debugging, and continuous improvement at scale** - 300+ peer code reviews enforcing documentation-based coding practices and engineering standards; reduced bug count to sub-10 for 6+ consecutive months; performance monitoring and ongoing maintenance across multi-tenant production environments.
- **Scalable, secure system design across the full stack** - Docker microservices redesign (50% scalability improvement, 30% deployment reduction); RabbitMQ event-driven processing; Redis caching; AWS S3, EC2, RDS; 200+ SQL query optimizations (30% gain); 100% PHI compliance with Libsodium-encrypted pipelines.
- Mentored 5+ engineers on OOP, code quality, and engineering best practices; collaborated with stakeholders to translate complex business problems into actionable technical requirements; participated in solution design evaluating trade-offs for scalable and maintainable systems.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built backend modules (PHP, Python, PostgreSQL) with OOP design patterns; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

## PROJECTS

### AgentProbe: Generative AI Evaluation & Quality Platform

GenAI · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Scalable generative AI integration platform: 15-endpoint FastAPI service with OOP-based architecture, 5-metric quality engine, CI-integrated regression gate, and multi-provider LLM routing. Demonstrates leveraging generative AI tools to enhance software quality at scale - zero regressions shipped across 1,240+ test cases.

### Leasel: 8-Agent Agentic AI & ML Integration System

AI/ML · Agentic

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI/ML integration: 8-agent LangGraph stateful system integrating machine learning components (TF-IDF, OCR, LLM orchestration) into a scalable, maintainable software platform with HITL escalation and CI quality gate. 93.6% precision across 1,240 compliance test cases.

### AI on FHIR: NLP-Driven Healthcare Query System

AI/ML · NLP

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI/ML integration: Python + React platform using BERT embeddings and LLM inference to enhance software capabilities with NLP - reduced retrieval time ~60%; HIPAA-compliant RBAC across structured medical records.

## EDUCATION

**M.S. Data Analytics Engineering -** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering -** JNTU Anantapur — 2013-2017